



5 Chebyshev Approximation

In this chapter, we study, for a compact domain $\Omega \subset \mathbb{R}^d$, $d \geq 1$, the approximation of continuous functions from the linear space

$$\mathcal{C}(\Omega) = \{u : \Omega \longrightarrow \mathbb{R} \mid u \text{ continuous}\}$$

with respect to the maximum norm

$$\|u\|_{\infty} = \max_{x \in \Omega} |u(x)| \quad \text{for } u \in \mathcal{C}(\Omega).$$

The maximum norm $\|\cdot\|_{\infty}$ is also referred to as *Chebyshev¹ norm*, and so in this chapter, we are concerned with *Chebyshev approximation*, i.e., approximation with respect to $\|\cdot\|_{\infty}$.

To approximate functions from $\mathcal{C}(\Omega)$, we work with *finite-dimensional* linear subspaces $\mathcal{S} \subset \mathcal{C}(\Omega)$. Under this assumption, there is for any $f \in \mathcal{C}(\Omega)$ a best approximation $s^* \in \mathcal{S}$ to f , see Corollary 3.8. Further in Chapter 3, we analyzed the problem of Chebyshev approximation from a more general viewpoint. Recall that we have made a negative observation: The Chebyshev norm $\|\cdot\|_{\infty}$ is *not* strictly convex, as shown in Example 3.34. According to Theorem 3.37, however, strictly convex norms guarantee (for convex $\mathcal{S} \subset \mathcal{F}$) the uniqueness of best approximations. Therefore, the problem of approximation with respect to the Chebyshev norm $\|\cdot\|_{\infty}$ appears to be, at first sight, rather critical.

But we should not be too pessimistic. In this chapter, we will derive suitable conditions on the approximation space $\mathcal{S} \subset \mathcal{C}(\Omega)$, under which we can even guarantee *strong* uniqueness of best approximations. However, according to the *Mairhuber-Curtis theorem*, Theorem 5.25, strong uniqueness can only be achieved for the univariate case. Therefore, the case $d = 1$, where $\Omega = [a, b] \subset \mathbb{R}$ is a compact interval, is of primary importance. In fact, we will study Chebyshev approximation to continuous functions in $\mathcal{C}[a, b]$ by algebraic polynomials from $\mathcal{P}_n$, for $n \in \mathbb{N}_0$, in more detail.

Furthermore, we derive suitable characterizations for best approximations for the particular case $\|\cdot\| = \|\cdot\|_{\infty}$, where we can rely on our previous results in Chapter 3. This finally leads us to the *Remez algorithm*, an iterative numerical method to compute best approximations with respect to the Chebyshev norm $\|\cdot\|_{\infty}$. We show linear convergence for the Remez iteration.

¹ PAFNUTY LVOVICH CHEBYSHEV (1821-1894), Russian mathematician

5.1 Approaches to Construct Best Approximations

For a compact domain $\Omega \subset \mathbb{R}^d$, $d \geq 1$, we denote by $\mathcal{C}(\Omega)$ the linear space of all continuous functions on Ω . Moreover, we assume throughout this chapter that $\mathcal{S} \subset \mathcal{C}(\Omega)$ is a finite-dimensional linear subspace of $\mathcal{C}(\Omega)$. Under these assumptions, there exists, according to Corollary 3.8, for any $f \in \mathcal{C}(\Omega)$ a best approximation $s^* \in \mathcal{S}$ to f with respect to the Chebyshev norm $\|\cdot\|_\infty$. However, s^* is not necessarily unique, since $\|\cdot\|_\infty$ is not strictly convex.

Now we apply the characterizations for best approximations from Chapter 3 to the special case of the Chebyshev norm $\|\cdot\|_\infty$. We begin with the direct characterizations from Section 3.4, where we had proven the Kolmogorov criterion, Corollary 3.55. We can adapt the Kolmogorov criterion to the Chebyshev norm $\|\cdot\|_\infty$ as follows.

Theorem 5.1. *Let $\mathcal{S} \subset \mathcal{C}(\Omega)$ be a linear subspace of $\mathcal{C}(\Omega)$ and suppose $f \in \mathcal{C}(\Omega) \setminus \mathcal{S}$. Then $s^* \in \mathcal{S}$ is a best approximation to f with respect to $\|\cdot\|_\infty$, if and only if*

$$\max_{x \in E_{s^*-f}} s(x) \operatorname{sgn}((s^* - f)(x)) \geq 0 \quad \text{for all } s \in \mathcal{S}, \quad (5.1)$$

where

$$E_{s^*-f} = \{x \in \Omega : |(s^* - f)(x)| = \|s^* - f\|_\infty\} \subset \Omega$$

denotes the set of **extremal points** of $s^* - f$ in Ω .

Proof. By the equivalence of the Kolmogorov criterion, Corollary 3.55, s^* is a best approximation to f with respect to $\|\cdot\| = \|\cdot\|_\infty$, if and only if

$$\|'_+(s^* - f, s - s^*) = \max_{x \in E_{s^*-f}} (s - s^*)(x) \operatorname{sgn}((s^* - f)(x)) \geq 0 \quad \text{for all } s \in \mathcal{S},$$

where we have used the Gâteaux derivative of the norm $\|\cdot\|_\infty$ from Theorem 3.64. By the linearity of $\mathcal{S}$, this condition is equivalent to (5.1). $\blacksquare$

Given the result of Theorem 5.1, we can immediately solve one simple problem of Chebyshev approximation. To this end, we regard the univariate case, $d = 1$, where $\Omega = [a, b] \subset \mathbb{R}$ for a compact interval. In this case, we wish to approximate continuous functions from $\mathcal{C}[a, b]$ by constants.

Corollary 5.2. *Let $[a, b] \subset \mathbb{R}$ be compact and $f \in \mathcal{C}[a, b]$. Then*

$$c^* = \frac{f_{\min} + f_{\max}}{2} \in \mathcal{P}_0$$

is the unique best approximation to f from $\mathcal{P}_0$ with respect to $\|\cdot\|_\infty$, where

$$f_{\min} = \min_{x \in [a, b]} f(x) \quad \text{and} \quad f_{\max} = \max_{x \in [a, b]} f(x).$$

Proof. For $f \in \mathcal{P}_0$, the statement is trivial. Now suppose $f \in \mathcal{C}[a, b] \setminus \mathcal{P}_0$. The continuous function $f \in \mathcal{C}[a, b]$ attains its minimum and maximum on the compact interval $[a, b]$. Therefore, there are $x_{\min}, x_{\max} \in [a, b]$ satisfying

$$f_{\min} = f(x_{\min}) \quad \text{and} \quad f_{\max} = f(x_{\max}).$$

Obviously, $x_{\min}, x_{\max}$ lie in the set of extremal points E_{c^*-f} , where

$$\begin{aligned} c^* - f(x_{\min}) &= \eta \\ c^* - f(x_{\max}) &= -\eta \end{aligned}$$

with $\eta = \|c^* - f\|_{\infty} = (f_{\max} - f_{\min})/2 > 0$. Moreover, in this case we have

$$\max_{x \in E_{c^*-f}} c \operatorname{sgn}(c^* - f(x)) = c \operatorname{sgn}(c^* - f(x_{\min})) = c \geq 0$$

for $c \geq 0$ on the one hand, and

$$\max_{x \in E_{c^*-f}} c \operatorname{sgn}(c^* - f(x)) = c \operatorname{sgn}(c^* - f(x_{\max})) = -c > 0$$

for $c < 0$ on the other hand. Altogether, the Kolmogorov criterion from Theorem 5.1,

$$\max_{x \in E_{c^*-f}} c \operatorname{sgn}(c^* - f(x)) \geq 0 \quad \text{for all } c \in \mathcal{P}_0,$$

is satisfied. Therefore, c^* is a best approximation to f from $\mathcal{P}_0$ with respect to $\|\cdot\|_{\infty}$.

Finally, c^* is the *unique* best approximation to f , since for $c \neq c^*$ we have

$$\begin{aligned} \|c - f\|_{\infty} &\geq |c - f_{\min}| > |c^* - f_{\min}| = \|c^* - f\|_{\infty} \quad \text{for } c > c^*; \\ \|c - f\|_{\infty} &\geq |c - f_{\max}| > |c^* - f_{\max}| = \|c^* - f\|_{\infty} \quad \text{for } c < c^*. \end{aligned}$$

■

Observe from the above construction of the unique best approximation $c^* \in \mathcal{P}_0$ to $f \in \mathcal{C}[a, b]$ that there are *at least* two different extremal points $x_1, x_2 \in E_{c^*-f}$ which satisfy the *alternation condition*

$$(c^* - f)(x_k) = (-1)^k \sigma \|c^* - f\|_{\infty} \quad \text{for } k = 1, 2 \quad (5.2)$$

for some $\sigma \in \{\pm 1\}$. The alternation condition (5.2) is necessary and sufficient for a best approximation from $\mathcal{P}_0$. Moreover, there is no upper bound for the number of alternation points. We can further explain this by the following simple example.

Example 5.3. We approximate $f_m(x) = \cos(mx)$, for $m \in \mathbb{N}$, on the interval $[-\pi, \pi]$ by constants. According to Corollary 5.2, $c^* \equiv 0$ is the unique best approximation to f_m from $\mathcal{P}_0$, for all $m \in \mathbb{N}$. We get $\|c^* - f_m\|_{\infty} = 1$ for the minimal distance between f_m and $\mathcal{P}_0$, and the error function $c^* - f_m$ has $2m + 1$ alternation points $x_k = \pi k/m$, for $k = -m, \dots, m$. ◇

For the approximation of $f \in \mathcal{C}[a, b]$ by polynomials from $\mathcal{P}_{n-1}$, there are *at least* $n+1$ alternation points. Moreover, the best approximation $p^* \in \mathcal{P}_{n-1}$ to f is unique. We can prove these two statements by another corollary from the Kolmogorov criterion, Theorem 5.1.

Corollary 5.4. *Let $[a, b] \subset \mathbb{R}$ be compact and $f \in \mathcal{C}[a, b] \setminus \mathcal{P}_{n-1}$, where $n \in \mathbb{N}$. Then there is a unique best approximation $p^* \in \mathcal{P}_{n-1}$ to f from $\mathcal{P}_{n-1}$ with respect to $\|\cdot\|_\infty$. Moreover, there are at least $n+1$ extremal points $\{x_1, \dots, x_{n+1}\} \subset E_{p^*-f}$ with $a \leq x_1 < \dots < x_{n+1} \leq b$, that are satisfying the alternation condition*

$$(p^* - f)(x_k) = (-1)^k \sigma \|p^* - f\|_\infty \quad \text{for } k = 1, \dots, n+1 \quad (5.3)$$

for some $\sigma \in \{\pm 1\}$.

Proof. The existence of a best approximation is covered by Corollary 3.8.

Now let $p^* \in \mathcal{P}_{n-1}$ be a best approximation to f . We decompose the set of extremal points E_{p^*-f} into m pairwise disjoint, non-empty, and monotonically increasing subsets, $E_1, \dots, E_m \subset E_{p^*-f}$, i.e.,

$$a \leq x_1 < x_2 < \dots < x_m \leq b \quad \text{for all } x_k \in E_k \text{ and } k = 1, \dots, m, \quad (5.4)$$

so that the sign of the error $p^* - f$ is alternating on the sets $E_k \subset E_{p^*-f}$, $1 \leq k \leq m$, i.e., we have, for some $\sigma \in \{\pm 1\}$,

$$\text{sgn}((p^* - f)(x_k)) = (-1)^k \sigma \quad \text{for all } x_k \in E_k \quad \text{for } k = 1, \dots, m. \quad (5.5)$$

We denote the order relation in (5.4) in short by $E_1 < \dots < E_m$.

Note that there are at least two extremal points in E_{p^*-f} , at which the error function $p^* - f$ has different signs. Indeed, this is because the continuous function $p^* - f$ attains its minimum and maximum on $[a, b]$, so that

$$(p^* - f)(x_{\min}) = -\|p^* - f\|_\infty \quad \text{and} \quad (p^* - f)(x_{\max}) = \|p^* - f\|_\infty$$

for $x_{\min}, x_{\max} \in [a, b]$. Otherwise, p^* cannot be a best approximation to f .

Therefore, there are at least two subsets E_k in the above decomposition of E_{p^*-f} , i.e., $m \geq 2$. We now show that we even have $m \geq n+1$ for the number of subsets E_k .

Suppose $m < n+1$, or, $m \leq n$. Then there is a set $X^* = \{x_1^*, \dots, x_{m-1}^*\}$ of size $m-1$, whose points x_k^* are located *between* the points from neighbouring subsets $E_k < E_{k+1}$, respectively, so that

$$x_k < x_k^* < x_{k+1} \quad \text{for all } x_k \in E_k, x_{k+1} \in E_{k+1} \text{ and } k = 1, \dots, m-1.$$

In this case, the corresponding knot polynomial

$$\omega_{X^*}(x) = \prod_{k=1}^{m-1} (x - x_k^*) \in \mathcal{P}_{m-1} \subset \mathcal{P}_{n-1}$$

has on the subsets E_k alternating signs, where

$$\operatorname{sgn}(\omega_{X^*}(x_k)) = (-1)^{m-k} \quad \text{for all } k = 1, \dots, m.$$

Now for the polynomial $p = p^* + \hat{\sigma} \omega_{X^*} \in \mathcal{P}_{n-1}$, with $\hat{\sigma} \in \{\pm 1\}$, we have

$$\operatorname{sgn}((p - p^*)(x_k)(p^* - f)(x_k)) = \hat{\sigma}(-1)^{m-k}(-1)^k \sigma = \hat{\sigma}(-1)^m \sigma$$

for all $x_k \in E_k$ and for $k = 1, \dots, m$. Letting $\hat{\sigma} = -(-1)^m \sigma$, we have

$$\max_{x_k \in E_{p^* - f}} (p - p^*)(x_k) \operatorname{sgn}((p^* - f)(x_k)) < 0.$$

This, however, is in contradiction to the Kolmogorov criterion, Theorem 5.1. Therefore, there are at least $m \geq n + 1$ monotonically increasing non-empty subsets $E_1 < \dots < E_m$ of $E_{p^* - f}$, for which the sign of the error function $p^* - f$ is alternating, i.e., we have (5.5) with $m \geq n + 1$, which implies the alternation condition (5.3).

Now we prove the uniqueness of p^* by contradiction.

Suppose there is another best approximation $q^* \in \mathcal{P}_{n-1}$ to f , $p^* \neq q^*$. Then, the convex combination $p = (p^* + q^*)/2 \in \mathcal{P}_{n-1}$ is according to Theorem 3.16 yet another best approximation to f . In this case, there are for p at least $n + 1$ alternation points $x_1 < \dots < x_{n+1}$, so that

$$(p - f)(x_k) = (-1)^k \sigma \|p - f\|_\infty \quad \text{for } k = 1, \dots, n + 1$$

for some $\sigma \in \{\pm 1\}$, where $\{x_1, \dots, x_{n+1}\} \subset E_{p-f}$.

But the $n + 1$ alternation points $x_1, \dots, x_{n+1}$ of p are also contained in each of the extremal point sets $E_{p^* - f}$ and $E_{q^* - f}$. Indeed, this is because in

$$\begin{aligned} \|p - f\|_\infty &= |(p - f)(x_k)| \leq \frac{1}{2}|(p^* - f)(x_k)| + \frac{1}{2}|(q^* - f)(x_k)| \\ &\leq \frac{1}{2}\|p^* - f\|_\infty + \frac{1}{2}\|q^* - f\|_\infty = \|p^* - f\|_\infty = \|q^* - f\|_\infty, \end{aligned}$$

equality holds for $k = 1, \dots, n + 1$. In particular, we have

$$|(p^* - f)(x_k) + (q^* - f)(x_k)| = |(p^* - f)(x_k)| + |(q^* - f)(x_k)|$$

for all $1 \leq k \leq n + 1$.

Due to the strict convexity of the norm $|\cdot|$ (see Remark 3.27) and by the equivalence statement (d) in Theorem 3.26, the signs of the error functions $p^* - f$ and $q^* - f$ must agree on $\{x_1, \dots, x_{n+1}\}$, i.e.,

$$\operatorname{sgn}((p^* - f)(x_k)) = \operatorname{sgn}((q^* - f)(x_k)) \quad \text{for all } k = 1, \dots, n + 1.$$

Altogether, the values of the polynomials $p^*, q^* \in \mathcal{P}_{n-1}$ coincide on the $n + 1$ points $x_1, \dots, x_{n+1}$, which implies $p^* \equiv q^*$. ■

Now we note another important corollary, which directly follows from our observation in Proposition 3.42 and from Exercise 3.73.

Corollary 5.5. *For $L > 0$ let $f \in \mathcal{C}[-L, L]$. Moreover, let $p^* \in \mathcal{P}_n$, for $n \in \mathbb{N}_0$, be the unique best approximation to f from $\mathcal{P}_n$ with respect to $\|\cdot\|_\infty$. Then the following statements are true.*

- (a) *If f is even, then its best approximation $p^* \in \mathcal{P}_n$ is even.*
- (b) *If f is odd, then its best approximation $p^* \in \mathcal{P}_n$ is odd.*

Proof. The linear space $\mathcal{P}_n$ of algebraic polynomials is *reflection-invariant*, i.e., for $p(x) \in \mathcal{P}_n$, we have $p(-x) \in \mathcal{P}_n$. Moreover, by Corollary 5.4 there exists for any $f \in \mathcal{C}[-L, L]$ a unique best approximation $p^* \in \mathcal{P}_n$ to f from $\mathcal{P}_n$ with respect to $\|\cdot\|_\infty$. Without loss of generality, we assume $L = 1$. By Proposition 3.42 and Exercise 3.73, both statements (a) and (b) hold. ■

For illustration, we apply Corollary 5.5 in the following two examples.

Example 5.6. We approximate $f_m(x) = \sin(mx)$, for $m \in \mathbb{N}$, on $[-\pi, \pi]$ by linear polynomials. The function f_m is odd, for all $m \in \mathbb{N}$, and so is the best approximation $p_m^* \in \mathcal{P}_1$ to f_m odd. Therefore, p_m^* has the form $p_m^*(x) = \alpha_m x$ for a slope $\alpha_m \geq 0$, which is yet to be determined.

Case 1: For $m = 1$, the constant $c \equiv 0$ cannot be a best approximation to $f_1(x) = \sin(x)$, since $c - f_1$ has only *two* alternation points $\pm\pi/2$. By symmetry, we can restrict our following investigations to the interval $[0, \pi]$. The function $p_1^*(x) - f_1(x) = \alpha_1 x - \sin(x)$, with $\alpha_1 > 0$, has two alternation points $\{x^*, \pi\}$ on $[0, \pi]$,

$$(p_1^* - f_1)(x^*) = \alpha_1 x^* - \sin(x^*) = -\eta \quad \text{and} \quad (p_1^* - f_1)(\pi) = \alpha_1 \pi = \eta,$$

where $\eta = \|p_1^* - f_1\|_\infty$ is the minimal distance between f_1 and $\mathcal{P}_1$. Moreover, the alternation point x^* satisfies the condition

$$0 = (p_1^* - f_1)'(x^*) = \alpha_1 - \cos(x^*) \quad \text{which implies } \alpha_1 = \cos(x^*).$$

Therefore, x^* is a solution of the nonlinear equation

$$\cos(x^*)(x^* + \pi) = \sin(x^*),$$

which we can solve numerically, whereby we obtain the alternation point $x^* \approx 1.3518$, the slope $\alpha_1 = \cos(x^*) \approx 0.2172$ and the minimal distance $\eta \approx 0.6825$. Altogether, the best approximation $p_1^*(x) = \alpha_1 x$ with $\{-\pi, -x^*, x^*, \pi\}$ gives *four* alternation points for $p_1^* - f_1$ on $[-\pi, \pi]$, see Figure 5.1 (a).

Case 2: For $m > 1$, $p_m^* \equiv 0$ is the unique best approximation to f_m .

For the minimal distance, we get $\|p_m^* - f_m\|_\infty = 1$ and the error function $p_m^* - f_m$ has $2m$ alternation points

$$x_k = \pm \frac{2k-1}{2m} \pi \quad \text{for } k = 1, 2, \dots, m,$$

see Figure 5.1 (b) for the case $m = 2$. ◇

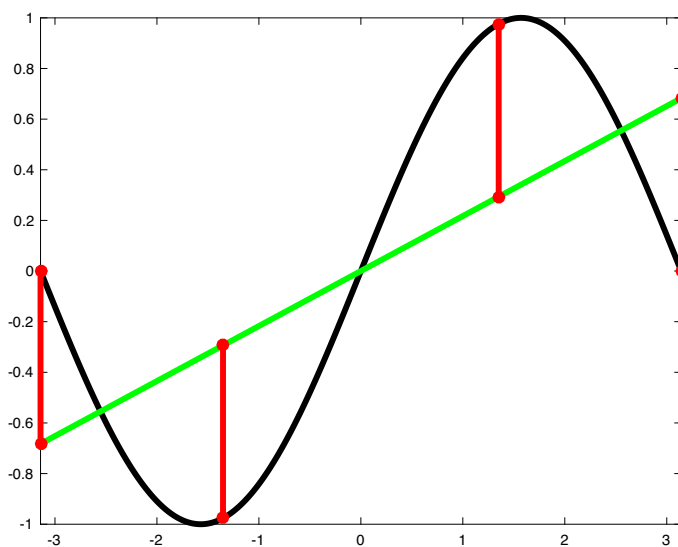
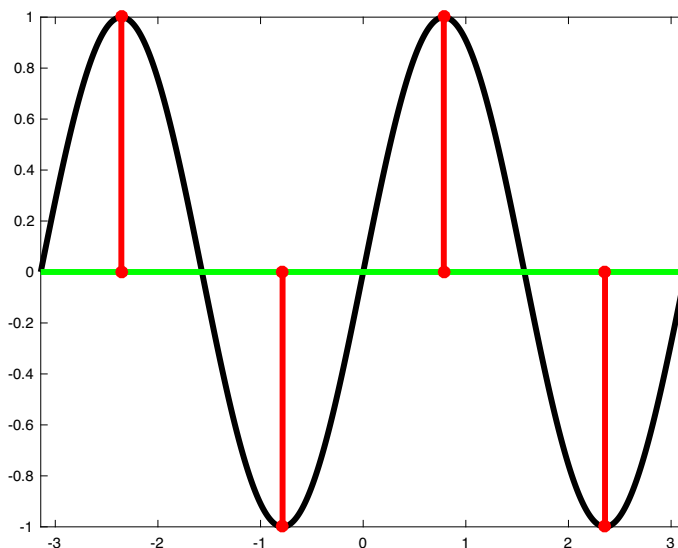
(a) approximation of the function $f_1(x) = \sin(x)$ (b) approximation of the function $f_2(x) = \sin(2x)$

Fig. 5.1. Approximation of the function $f_m(x) = \sin(mx)$ on $[-\pi, \pi]$ by linear polynomials for (a) $m = 1$ and (b) $m = 2$. The best approximation $p_m^* \in \mathcal{P}_1$ to f_m , $m = 1, 2$, is odd. In Example 5.6, we determine the best approximation $p_m^* \in \mathcal{P}_1$ to f_m and the corresponding alternation points for all $m \in \mathbb{N}$.

Regrettably, the characterization of best approximations in Corollary 5.4 is not constructive, since neither do we know the set of extremal points E_{p^*-f} nor do we know the minimal distance $\|p^* - f\|_\infty$ a priori. Otherwise, we could immediately compute the best approximating polynomial $p^* \in \mathcal{P}_{n-1}$ from the interpolation conditions

$$p^*(x_k) = f(x_k) + (-1)^k \eta \quad \text{where } \eta = \sigma \|p^* - f\|_\infty,$$

for $k = 1, \dots, n+1$. For further illustration, we discuss the following example, where we can predetermine some of the extremal points.

Example 5.7. We approximate the absolute-value function $f(x) = |x|$ on $[-1, 1]$ by quadratic polynomials. To construct the best approximation $p_2^* \in \mathcal{P}_2$ to $f \in \mathcal{C}[-1, 1]$ we first note the following observations.

- The function f is even, and so p_2^* must be even, by Corollary 5.5.
- By Corollary 5.4 there are *at least* four extremal points, $|E_{p_2^*-f}| \geq 4$.
- The error function $e = p_2^* - f$ on $[0, 1]$ is a quadratic polynomial. Therefore, e has on $(0, 1)$ at most one local extremum $x^* \in (0, 1)$. This local extremum must lie in the set of extremal points $E_{p_2^*-f}$. By symmetry, $-x^* \in (-1, 0)$ is also contained in the set of extremal points $E_{p_2^*-f}$.
- Further extrema of the error function e can only be at the origin or at the boundary points ± 1 . Since $|E_{p_2^*-f}| \geq 4$ and due to symmetry, *both* boundary points ± 1 must lie in $E_{p_2^*-f}$.
- To satisfy the alternation condition, the origin must necessarily lie in $E_{p_2^*-f}$. Indeed, for the subset $E = \{-1, -x^*, x^*, 1\} \subset E_{p_2^*-f}$ the four signs of e on E are symmetric, in particular *not* alternating. Therefore, we have $E_{p_2^*-f} = \{-1, -x^*, 0, x^*, 1\}$ for some $x^* \in (0, 1)$.
- By symmetry we can restrict ourselves in the following investigations to the unit interval $[0, 1]$: Since the error function $e = p_2^* - f$ has three extrema $\{0, x^*, 1\}$ in $[0, 1]$, e has two zeros in $(0, 1)$, i.e., the function graphs of f and p_2^* intersect in $(0, 1)$ at two points. Hence, p_2^* is convex, where $p_2^*(0) > 0$.

We can now sketch the function graphs of f and p_2^* (see Figure 5.2).

By our above observations the best approximation p_2^* has the form

$$p_2^*(x) = \eta + \alpha x^2$$

with the minimal distance $\eta = \|p_2^* - f\|_\infty$, for some positive slope $\alpha > 0$. Moreover, $e = p_2^* - f$ has on the set of extremal points $E_{p_2^*-f} = \{-1, -x^*, 0, x^*, 1\}$ alternating signs $\varepsilon = (1, -1, 1, -1, 1)$. We compute α by the alternation condition at $x = 1$,

$$(p_2^* - f)(1) = \eta + \alpha - 1 = \eta,$$

and so we obtain $\alpha = 1$, so that $p_2^*(x) = \eta + x^2$. The local minimum x^* of the error function $e = p_2^* - f$ satisfies the necessary condition

$$e'(x^*) = 2x^* - 1 = 0,$$

whereby $x^* = 1/2$, so that $E_{p_2^*-f} = \{-1, -1/2, 0, 1/2, 1\}$. Finally, at $x^* = 1/2$ we have the alternation condition

$$(p_2^* - f)(1/2) = \eta + 1/4 - 1/2 = -\eta,$$

holds, whereby $\eta = 1/8$. Hence, the quadratic polynomial $p_2^*(x) = 1/8 + x^2$ is the unique best approximation to f from $\mathcal{P}_2$ with respect to $\|\cdot\|_\infty$. $\diamond$

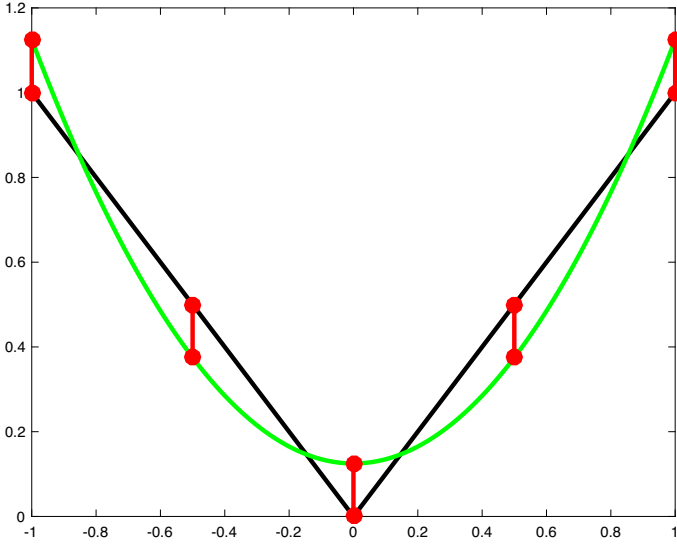


Fig. 5.2. Approximation of the function $f(x) = |x|$ on $[-1, 1]$ by quadratic polynomials. The best approximation $p_2^* \in \mathcal{P}_2$ to f is even and convex. The set of extremal points $E_{p_2^*-f} = \{-1, -x^*, 0, x^*, 1\}$ has five alternation points.

A more constructive account for the computation of best approximations relies on the dual characterizations from Section 3.3. To this end, we recall the necessary and sufficient condition from Theorem 3.48. According to Theorem 3.48, $s^* \in \mathcal{S} \subset \mathcal{C}(\Omega)$ is a best approximation to $f \in \mathcal{C}(\Omega)$ with respect to $\|\cdot\|_\infty$, if and only if there is a dual functional $\varphi \in (\mathcal{C}(\Omega))'$ satisfying

- (a) $\|\varphi\|_\infty = 1$.
- (b) $\varphi(s^* - f) = \|s^* - f\|_\infty$.
- (c) $\varphi(s - s^*) \geq 0$ for all $s \in \mathcal{S}$.

To construct such a characterizing dual functional, we use the assumption

$$\varphi(u) = \sum_{k=1}^m \lambda_k \varepsilon_k u(x_k) \quad \text{for } u \in \mathcal{C}(\Omega) \quad (5.6)$$

with coefficient vector $\lambda = (\lambda_1, \dots, \lambda_m)^T \in A_m$, lying at the boundary

$$A_m = \left\{ (\lambda_1, \dots, \lambda_m)^T \in \mathbb{R}^m \mid \lambda_k \in [0, 1], 1 \leq k \leq m, \sum_{k=1}^m \lambda_k = 1 \right\} \quad (5.7)$$

of the standard simplex $\Delta_m \subset \mathbb{R}^m$ from (2.38). Moreover,

$$\varepsilon = (\varepsilon_1, \dots, \varepsilon_m)^T \in \{\pm 1\}^m$$

denotes a sign vector and $X = \{x_1, \dots, x_m\} \subset \Omega$ is a point set.

Assuming (5.6) condition (a) is already satisfied, since

$$|\varphi(u)| = \left| \sum_{k=1}^m \lambda_k \varepsilon_k u(x_k) \right| \leq \|u\|_\infty \quad \text{for all } u \in \mathcal{C}(\Omega) \quad (5.8)$$

and so $\|\varphi\|_\infty \leq 1$. Moreover, for any $u \in \mathcal{C}(\Omega)$ satisfying $\|u\|_\infty = 1$ and $u(x_k) = \varepsilon_k$, for all $1 \leq k \leq m$, we have equality in (5.8), so that φ has norm length one by $\|\varphi\|_\infty = \|\lambda\|_1 = 1$.

To satisfy condition (b), we choose $X = E_{s^*-f}$, i.e., $E_{s^*-f} = \{x_1, \dots, x_m\}$. In this case, we get, in combination with $\varepsilon_k = \text{sgn}((s^* - f)(x_k))$, the identity

$$\varphi(s^* - f) = \sum_{k=1}^m \lambda_k \varepsilon_k (s^* - f)(x_k) = \sum_{k=1}^m \lambda_k |(s^* - f)(x_k)| = \|s^* - f\|_\infty.$$

But the set of extremal points E_{s^*-f} is unknown a priori. Moreover, it remains to satisfy condition (c).

From now on we study the construction of coefficients $\lambda \in A_m$, signs $\varepsilon \in \{\pm 1\}^m$ and points $X = \{x_1, \dots, x_m\}$ in more detail. In order to do so, we need some technical preparations. We begin with the representation of convex hulls.

Definition 5.8. Let $\mathcal{F}$ be a linear space and $\mathcal{M} \subset \mathcal{F}$. Then the **convex hull** $\text{conv}(\mathcal{M})$ of $\mathcal{M}$ is the smallest convex set in $\mathcal{F}$ containing $\mathcal{M}$, i.e.,

$$\text{conv}(\mathcal{M}) = \bigcap_{\substack{\mathcal{M} \subset \mathcal{K} \subset \mathcal{F} \\ \mathcal{K} \text{ convex}}} \mathcal{K}.$$

○

The following representation for $\text{conv}(\mathcal{M})$ is much more useful in practice.

Theorem 5.9. *Let $\mathcal{F}$ be a linear space and $\mathcal{M} \subset \mathcal{F}$. Then we have*

$$\text{conv}(\mathcal{M}) = \left\{ \sum_{j=1}^m \lambda_j x_j \mid x_j \in \mathcal{M} \text{ and } \lambda = (\lambda_1, \dots, \lambda_m)^T \in \Lambda_m \text{ for } m \in \mathbb{N} \right\}.$$

Proof. Let us consider the set

$$\mathcal{K} = \left\{ \sum_{j=1}^m \lambda_j x_j \mid x_j \in \mathcal{M} \text{ and } \lambda = (\lambda_1, \dots, \lambda_m)^T \in \Lambda_m \text{ for } m \in \mathbb{N} \right\}. \quad (5.9)$$

We now show the following properties for $\mathcal{K}$.

- (a) $\mathcal{K}$ is convex.
- (b) $\mathcal{M} \subset \mathcal{K}$.
- (c) $\text{conv}(\mathcal{M}) \subset \mathcal{K}$.

(a): For $x, y \in \mathcal{K}$ we have the representations

$$x = \sum_{j=1}^m \lambda_j x_j \quad \text{with } \lambda = (\lambda_1, \dots, \lambda_m)^T \in \Lambda_m, \{x_1, \dots, x_m\} \subset \mathcal{M}, m \in \mathbb{N}$$

$$y = \sum_{k=1}^n \mu_k y_k \quad \text{with } \mu = (\mu_1, \dots, \mu_n)^T \in \Lambda_n, \{y_1, \dots, y_n\} \subset \mathcal{M}, n \in \mathbb{N}.$$

Note that any convex combination $\alpha x + (1 - \alpha)y$, $\alpha \in [0, 1]$, can be written as a convex combination of the points $x_1, \dots, x_m, y_1, \dots, y_n$,

$$\alpha x + (1 - \alpha)y = \alpha \sum_{j=1}^m \lambda_j x_j + (1 - \alpha) \sum_{k=1}^n \mu_k y_k = \sum_{j=1}^m \alpha \lambda_j x_j + \sum_{k=1}^n (1 - \alpha) \mu_k y_k,$$

so that $\alpha x + (1 - \alpha)y \in \mathcal{K}$ for all $\alpha \in [0, 1]$.

(b): Any point $x \in \mathcal{M}$ lies in $\mathcal{K}$, by $m = 1$, $\lambda_1 = 1$ and $x_1 = x$ in (5.9). Therefore, the inclusion $\mathcal{M} \subset \mathcal{K}$ holds.

(c): By (a) and (b) $\mathcal{K}$ is a convex set containing $\mathcal{M}$. From the minimality of $\text{conv}(\mathcal{M})$ we can conclude $\text{conv}(\mathcal{M}) \subset \mathcal{K}$.

We now show the inclusion $\mathcal{K} \subset \text{conv}(\mathcal{M})$. To this end, we first note that any convex $\mathcal{L}$ containing $\mathcal{M}$, i.e., $\mathcal{M} \subset \mathcal{L}$, is necessarily a superset of $\mathcal{K}$, i.e., $\mathcal{K} \subset \mathcal{L}$. Indeed, this is because $\mathcal{L}$ contains all finite convex combinations of points from $\mathcal{M}$. This immediately implies

$$\mathcal{K} \subset \bigcap_{\substack{\mathcal{M} \subset \mathcal{L} \\ \mathcal{L} \text{ convex}}} \mathcal{L} = \text{conv}(\mathcal{M}).$$

Altogether, we have $\mathcal{K} = \text{conv}(\mathcal{M})$. ■

By the characterization in Theorem 5.9, we can identify the convex hull $\text{conv}(\mathcal{M})$, for any set $\mathcal{M} \subset \mathcal{F}$, as the set of all *finite* convex combinations of points from $\mathcal{M}$. For *finite-dimensional* linear spaces $\mathcal{F}$, the length of the convex combinations can uniformly be bounded above according to the *Carathéodory² theorem*.

Theorem 5.10. (Carathéodory).

Let $\mathcal{F}$ be a linear space of finite dimension $\dim(\mathcal{F}) = n < \infty$. Moreover, suppose $\mathcal{M} \subset \mathcal{F}$. Then we have the representation

$$\text{conv}(\mathcal{M}) = \left\{ \sum_{j=1}^m \lambda_j x_j \mid x_j \in \mathcal{M}, \lambda = (\lambda_1, \dots, \lambda_m)^T \in \Lambda_m \text{ for } m \leq n+1 \right\}.$$

Proof. For $x \in \text{conv}(\mathcal{M})$ we consider a representation of the form

$$x = \sum_{j=1}^m \lambda_j x_j \quad \text{with } \lambda = (\lambda_1, \dots, \lambda_m)^T \in \Lambda_m \text{ and } x_1, \dots, x_m \in \mathcal{M}$$

but with *minimal* $m \in \mathbb{N}$. Then $\lambda_j > 0$, i.e., $\lambda_j \in (0, 1]$ for all $1 \leq j \leq m$. From the assumed representation we get

$$\sum_{j=1}^m \lambda_j (x - x_j) = 0,$$

i.e., the elements $x - x_j \in \mathcal{F}$, $1 \leq j \leq m$, are linearly dependent in $\mathcal{F}$.

Now suppose $m > n+1$, or, $m-1 > n$. Then there are $\alpha_2, \dots, \alpha_m \in \mathbb{R}$, that are not all vanishing, with

$$\sum_{j=1}^m \alpha_j (x - x_j) = 0,$$

where we let $\alpha_1 = 0$. This gives the representation

$$0 = \sum_{j=1}^m (\lambda_j + t\alpha_j)(x - x_j) = \sum_{j=1}^m \mu_j(t)(x - x_j) \quad \text{for all } t \in \mathbb{R},$$

with $\mu_j(t) = (\lambda_j + t\alpha_j)$ and therefore $\mu_j(0) = \lambda_j > 0$.

Now we choose one $t^* \in \mathbb{R}$ satisfying

$$\mu_j(t^*) = \lambda_j + t^* \alpha_j \geq 0 \quad \text{for all } j = 1, \dots, m,$$

and $\mu_k(t^*) = 0$ for some $k \in \{1, \dots, m\}$. By

² CONSTANTIN CARATHÉODORY (1873-1950), Greek mathematician

$$\rho_j = \frac{\mu_j(t^*)}{\sum_{k=1}^m \mu_k(t^*)} \geq 0 \quad \text{for } j = 1, \dots, m$$

we have

$$\sum_{j=1}^m \rho_j = 1$$

and

$$\sum_{j=1}^m \rho_j (x - x_j) = 0 \quad \Longleftrightarrow \quad x = \sum_{j=1}^m \rho_j x_j.$$

Note that $\rho_k = 0$. But this is in contradiction to the minimality of m . ■

The Carathéodory theorem implies the following important result.

Corollary 5.11. *Let $\mathcal{F}$ be a normed linear space of finite dimension $n < \infty$. Suppose $\mathcal{M} \subset \mathcal{F}$ is a compact subset of $\mathcal{F}$. Then $\text{conv}(\mathcal{M})$ is compact.*

Proof. We regard on the compact set $\mathcal{L}_{n+1} = \Lambda_{n+1} \times \mathcal{M}^{n+1}$ the continuous mapping $\varphi : \mathcal{L}_{n+1} \rightarrow \mathcal{F}$, defined as

$$\varphi(\lambda, X) = \sum_{j=1}^{n+1} \lambda_j x_j$$

for $\lambda = (\lambda_1, \dots, \lambda_{n+1})^T \in \Lambda_{n+1}$ and $X = (x_1, \dots, x_{n+1}) \in \mathcal{M}^{n+1}$. According to the Carathéodory theorem, Theorem 5.10, we have $\varphi(\mathcal{L}_{n+1}) = \text{conv}(\mathcal{M})$. Therefore, $\text{conv}(\mathcal{M})$ is also compact, since $\text{conv}(\mathcal{M})$ is the image of the compact set $\mathcal{L}_{n+1}$ under the continuous mapping $\varphi : \mathcal{L}_{n+1} \rightarrow \mathcal{F}$. ■

From Corollary 5.11, we gain the following separation theorem.

Theorem 5.12. *Let $\mathcal{M} \subset \mathbb{R}^d$ be compact. Then the following two statements are equivalent.*

- (a) *There is no $\beta \in \mathbb{R}^d \setminus \{0\}$ satisfying $\beta^T x > 0$ for all $x \in \mathcal{M}$.*
- (b) *$0 \in \text{conv}(\mathcal{M})$.*

Proof. (b) $\Rightarrow$ (a): Let $0 \in \text{conv}(\mathcal{M})$. Then we have the representation

$$0 = \sum_{j=1}^m \lambda_j x_j \quad \text{with } \lambda = (\lambda_1, \dots, \lambda_m)^T \in \Lambda_m \text{ and } x_1, \dots, x_m \in \mathcal{M}.$$

Suppose there is one $\beta \in \mathbb{R}^d \setminus \{0\}$ satisfying $\beta^T x > 0$ for all $x \in \mathcal{M}$. Then we immediately get a contradiction by

$$\beta^T 0 = 0 = \sum_{j=1}^m \lambda_j \beta^T x_j > 0.$$

(a) $\Rightarrow$ (b): Suppose statement (a) holds. Further suppose that $0 \notin \text{conv}(\mathcal{M})$.

Since $\text{conv}(\mathcal{M})$ is compact, by Corollary 5.11, there is one $\beta_* \in \text{conv}(\mathcal{M})$, $\beta_* \neq 0$, of minimal Euclidean norm in $\text{conv}(\mathcal{M})$. This minimum β_* , viewed as a best approximation from $\text{conv}(\mathcal{M})$ to the origin with respect to $\|\cdot\|_2$, is characterized by

$$(\beta_* - 0, x - \beta_*) \geq 0 \quad \text{for all } x \in \text{conv}(\mathcal{M})$$

according to the Kolmogorov theorem, Corollary 3.55, in combination with the Gâteaux derivative for Euclidean norms in Theorem 3.62. But this condition is equivalent to

$$\beta_*^T x = (\beta_*, x) \geq (\beta_*, \beta_*) = \|\beta_*\|_2^2 > 0 \quad \text{for all } x \in \text{conv}(\mathcal{M}).$$

But this is in contradiction to our assumption in (a). ■

Remark 5.13. The equivalence statement (a) in Theorem 5.12 says that the Euclidean space $\mathbb{R}^d$ cannot be split by a separating hyperplane through the origin into two half-spaces, such that $\mathcal{M}$ is entirely contained in one of the two half-spaces. □

5.2 Strongly Unique Best Approximations

Now we wish to further develop the characterizations for best approximations from Sections 3.3 and 3.4 for the special case of the Chebyshev norm $\|\cdot\|_\infty$. In the following discussion, $\{s_1, \dots, s_n\} \subset \mathcal{S}$, for $n \in \mathbb{N}$, denotes a basis for the finite-dimensional approximation space $\mathcal{S} \subset \mathcal{C}(\Omega)$. To characterize a best approximation $s^* \in \mathcal{S}$ to some $f \in \mathcal{C}(\Omega) \setminus \mathcal{S}$, we work with the compact point set

$$\mathcal{M}_{s^*-f} = \{(s^* - f)(x)(s_1(x), \dots, s_n(x))^T \mid x \in E_{s^*-f}\} \subset \mathbb{R}^n,$$

where we can immediately draw the following conclusion from Theorem 5.12.

Corollary 5.14. *For $s^* \in \mathcal{S}$ the following statements are equivalent.*

- (a) s^* is a best approximation to $f \in \mathcal{C}(\Omega) \setminus \mathcal{S}$.
- (b) $0 \in \text{conv}(\mathcal{M}_{s^*-f})$.

Proof. In this proof, we use the notation

$$s_\beta = \sum_{j=1}^n \beta_j s_j \in \mathcal{S} \quad \text{for } \beta = (\beta_1, \dots, \beta_n)^T \in \mathbb{R}^n. \quad (5.10)$$

(b) $\Rightarrow$ (a): Let $0 \in \text{conv}(\mathcal{M}_{s^*-f})$. Suppose $s^* \in \mathcal{S}$ is *not* a best approximation to f . Then there is one $\beta = (\beta_1, \dots, \beta_n)^T \in \mathbb{R}^n \setminus \{0\}$ satisfying

$$\|s^* - f - s_\beta\|_\infty < \|s^* - f\|_\infty$$